



UNIVERSITY
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Statistical Analysis of Online Chess

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1. Data Description

1.1 Introduction

Chess is a well-known game that depends on tactics, wisdom, and decision-making, and with the rise of online platforms like Lichess and millions of games are played, and these games capture information on players, moves, and results. Analyzing this data can provide insights into how factors such as player rating, choice of moves, and the time taken to make a move influence game outcomes.

1.2 Literature Review

Chess Openings' Complexity, and Their Effect on Success:

The authors of Quantifying the complexity and similarity of chess openings using online chess community data analyze a massive dataset of online games to map a “relatedness network” of openings, clustering them by similarity and measuring each opening’s “complexity.” They find that weaker players concentrate on low-complexity, popular openings, while stronger players more frequently adopt harder, less common openings. The study links players’ “complex openings” with higher overall ratings, suggesting that opening choice and complexity affect win probability — but through a nuanced interplay of player skill, preparation, and opening difficulty (De Marzo & Servedio, 2023).

Time Pressure and Game Outcome in Chess:

The study The effects of time pressure on chess skill: an investigation into fast and slow processes underlying expert performance (van Harreveld, Wagenmakers & van der Maas, 2007) investigates how tightening time controls changes the predictive power of player skill on game outcome. Across both online and world-championship games, they observe that as time becomes scarce, differences in player skill matters less — suggesting that fast pattern-recognition processes dominate under pressure, while deep calculation (slow processes) becomes less feasible. As a result, games under severe time constraints become more random and less determined by rating differences, reducing the advantage of higher-rated players (van Harreveld et al., 2007)

1.3 Questions that can be answered by the analysis of data

- Does a player who has a higher rating always win the game?
- Who wins more, the black or the white?
- What is the victory status that is used more often? (checkmate, resignation, etc...)

1.4 The sources

The “Chess Game Dataset (Lichess)” is available at: <https://www.kaggle.com/datasets/datasnaek/chess/data>; it contains more than 20,000 games played on Lichess. Each observation corresponds to a single chess game played online on Lichess. There are over 20,000 such observations, and these observations are summarized in an Excel sheet that will be attached to the submission of the project.

Variables Description Table

#	Variable Name	Type	Level of Measurement	Description	Unit of Measurement
1	id	Qualitative	Nominal	A unique identifier for every chess game.	None
2	rated	Qualitative	Nominal (Binary)	Indicates if the game is rated or not.	None
3	turns	Quantitative	Discrete	The total number of moves in the game.	Number of turns
4	victory_status	Qualitative	Nominal	how the game ended (checkmate, resignation)	None
5	winner	Qualitative	Nominal	Indicates who won (White, Black, or Draw).	None
6	increment_code	Qualitative	Nominal	Describes the time control format (e.g., "3+2")	Minutes. + Seconds
7	white_id	Qualitative	Nominal	The username or ID of the white player.	None
8	white_rating	Quantitative	Discrete	The player's skill rating before the game.	Rating points
9	black_id	Qualitative	Nominal	The username or ID of the black player.	None
10	black_rating	Quantitative	Discrete	The player's skill rating before the game.	Rating points
11	moves	Qualitative	Nominal	The sequence of moves made during the game.	None
12	opening_name	Qualitative	Nominal	The name of the chess opening used	None.

Table. 1: Variables Description

2. Descriptive statistics

2.1 Rated players

Category	Frequency (Count)	Relative Frequency (%)
TRUE (Rated)	16,155	80.54%
FALSE (Unrated)	3,903	19.46%
Total	20,058	100%

Table 2: Rated players

2.2 Turns

Variable	Mean	Median	Mode	SD	Min	Max	Shape
Turns	60.5	55.0	53	33.6	1	349	Right Skewed

Table 3: Turns

2.3 victory_status

Category	Frequency (Count)	Relative Frequency (%)
Resign	11,147	55.57%
Mate	6,325	31.53%
Out of Time	1,680	8.38%
Draw	906	4.52%
Total	20,058	100%

Table 4: victory_status

2.4 winner

Winner	Frequency (Count)	Relative Frequency (%)
White	10,001	49.86%
Black	9,107	45.40%
Draw	950	4.74%
Total	20,058	100%

Table 5: Winner

2.5 Increment_code

Increment Code	Frequency (Count)	Relative Frequency (%)
10+0	7,721	38.49%
15+0	1,311	6.54%
15+15	850	4.24%
5+5	738	3.68%
5+8	697	3.47%
8+0	588	2.93%
10+5	579	2.89%
15+10	461	2.30%
20+0	448	2.23%
30+0	375	1.87%
10+10	355	1.77%
15+5	314	1.57%
7+2	265	1.32%
10+2	187	0.93%
30+30	169	0.84%
... (Other 385 codes)	4,990	24.88%
Total	20,058	100%

Table 6: Increment code

2.6 White rating

Variable	Mean	Median	Mode	SD	Min	Max	Shape
white_rating	1597	1567	1500	291	784	2700	Right Skewed

Table 7: White rating

2.7 Black rating

Variable	Mean	Median	Mode	SD	Min	Max	Shape
black_rating	1589	1562	1500	291	789	2723	Right Skewed

Table 8: Black rating

2.8 Opening names

Opening Name	Frequency (Count)	Relative Frequency (%)
Van't Kruijs Opening	368	1.83%
Sicilian Defense	358	1.78%
Sicilian Defense: Bowdler Attack	296	1.48%
Scotch Game	271	1.35%
French Defense: Knight Variation	271	1.35%
Scandinavian Defense: Mieses-Kotroc Variation	259	1.29%
Queen's Pawn Game: Mason Attack	232	1.16%
Queen's Pawn Game: Chigorin Variation	229	1.14%
Scandinavian Defense	223	1.11%
Horwitz Defense	209	1.04%
Caro-Kann Defense	199	0.99%
Philidor Defense #3	198	0.99%
Philidor Defense #2	193	0.96%
Indian Game	181	0.90%
Italian Game: Anti-Fried Liver Defense	180	0.90%
... (Other 1,462 openings)	...	...
Total	20,058	100%

Table 9: Opening names (Top 15)

3. Data visualization

3.1 Rated players

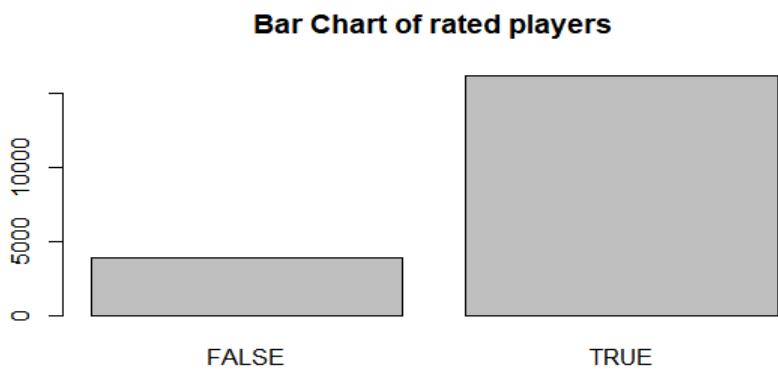


Figure 1: Bar Chart of Rated vs. Unrated Games

Comment: The bar chart clearly shows that the vast majority of games in this dataset are rated. Specifically, 80.5% (16,155 games) were played for a rating, while only 19.5% (3,903 games) were casual (unrated) matches.

Pie Chart of rated players

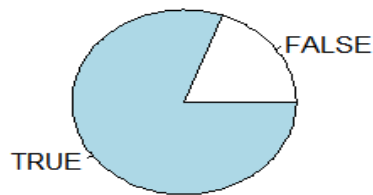


Figure 2: Pie Chart of Rated vs. Unrated Games

Comment: The pie chart visually demonstrates that rated games dominate the dataset, occupying more than three-quarters of the chart (80.5%). The smaller slice representing Unrated games (19.5%) indicates that casual play is much less common than competitive play on this platform.

3.2 Turns

Histogram of turns

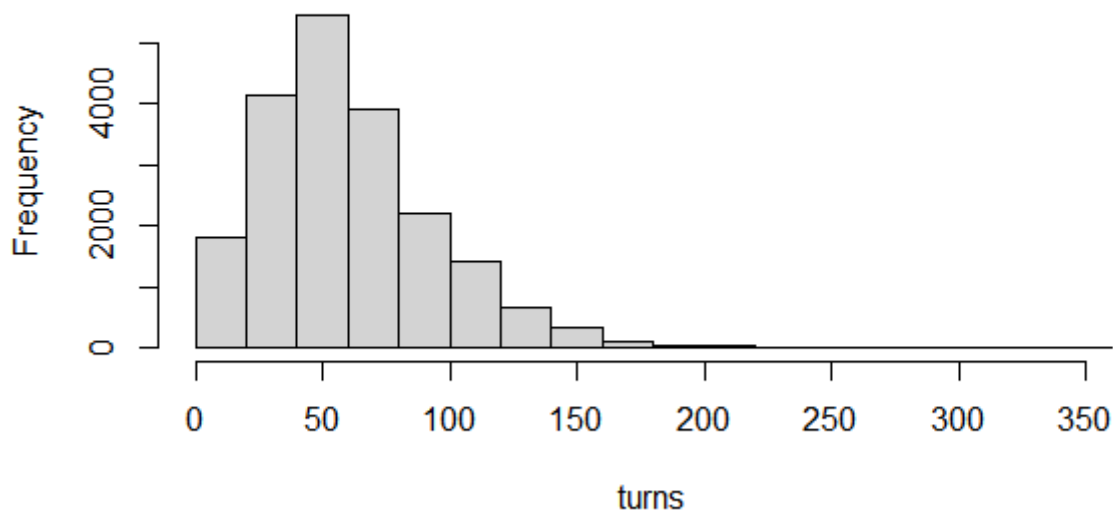


Figure 3: Histogram of Turns

Comment: The histogram is clearly right-skewed, meaning that most chess games are relatively short (peaking around 40-60 moves). However, there is a long tail extending to the right, representing the rare marathon games that last over 150 or 200 moves.

boxplot of Turns

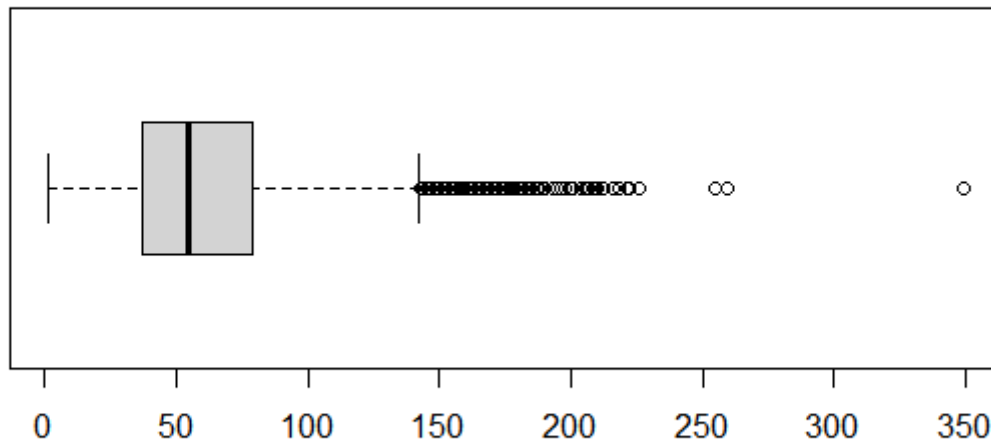


Figure 4: Boxplot of Turns

Comment: The boxplot shows a median game length of 55 turns. The "box" (Interquartile Range) is relatively tight, but there are a significant number of outliers on the upper end. This confirms that while most games finish within a standard range, there are exceptional cases where games continue for a very long time.

Ogive Plot for turns

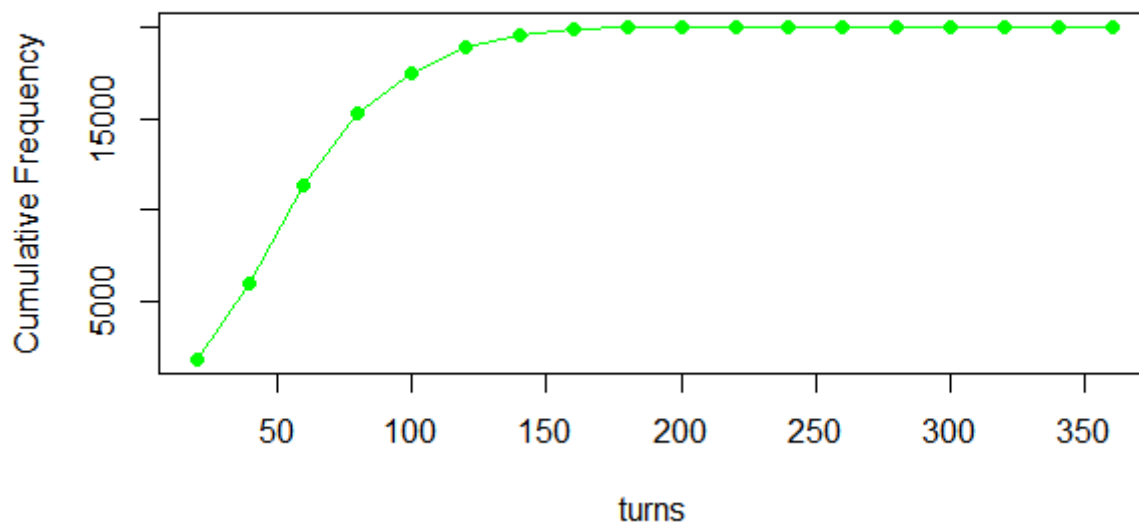


Figure 5: Ogive of Turns

Comment: The steep slope at the beginning of the Ogive indicates that a large percentage of games are completed quickly (under 60 turns). The curve then flattens out as the number of turns increases, showing that fewer and fewer games reach those high turn counts.

3.3 Victory Status

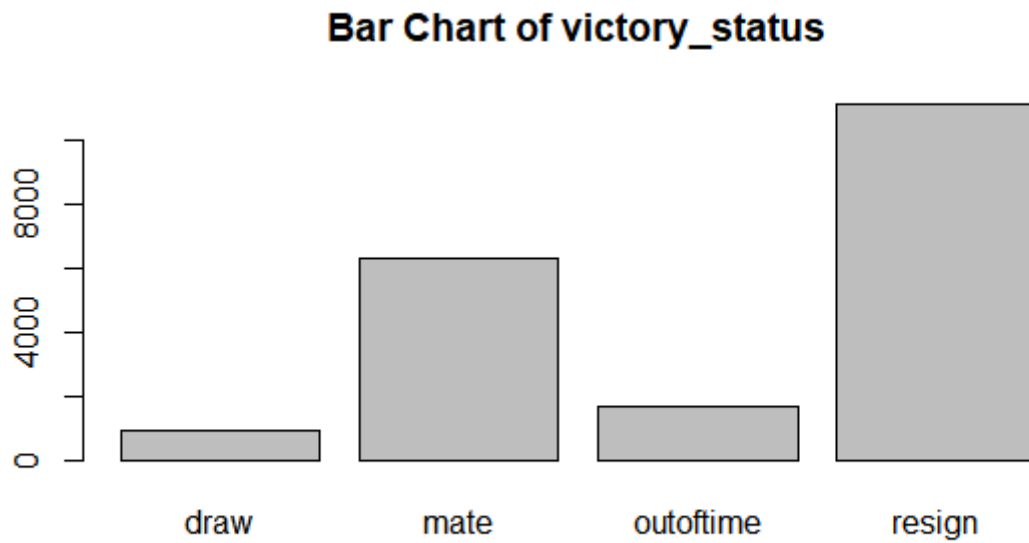


Figure 6: Bar Chart of Victory Status

Comment: The bar chart shows that Resignation is by far the most common way games end, accounting for over half of all matches (55.6%). Checkmates are the second most common result (31.5%), while games ending by time expiration or a draw are relatively rare.

Pie Chart of victory_status

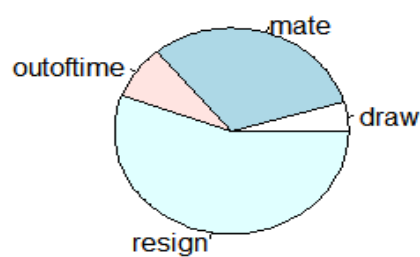


Figure 7: Pie Chart of Victory Status

Comment: The pie chart illustrates the dominance of decisive results, with the "Resign" and "Mate" slices combined making up 87% of the chart. The small slice for "Draw" (4.5%) highlights that tie games are an uncommon outcome in this dataset.

3.4 winner

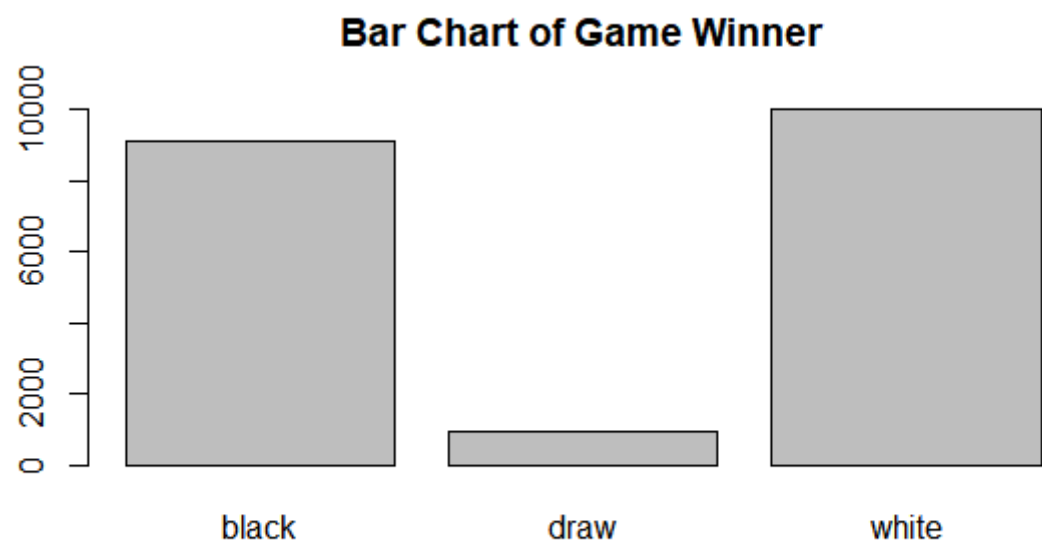


Figure 8: Bar Chart of Game Outcomes

Comment: The bar chart illustrates that White wins slightly more frequently than Black, with White securing 49.9% of victories compared to Black's 45.4%. Draws are the least common outcome, occurring in only 4.7% of the games.

Pie Chart of Game Winner

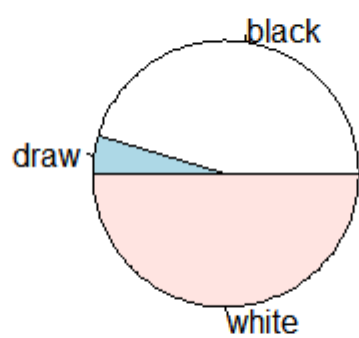


Figure 9: Pie Chart of Game Outcomes

Comment: The pie chart visually confirms that decisive results (wins for either White or Black) dominate the dataset. The slices for White and Black are nearly equal in size, while the small slice for Draws highlights that tie games are relatively rare in this level of play.

3.5 Increment_code

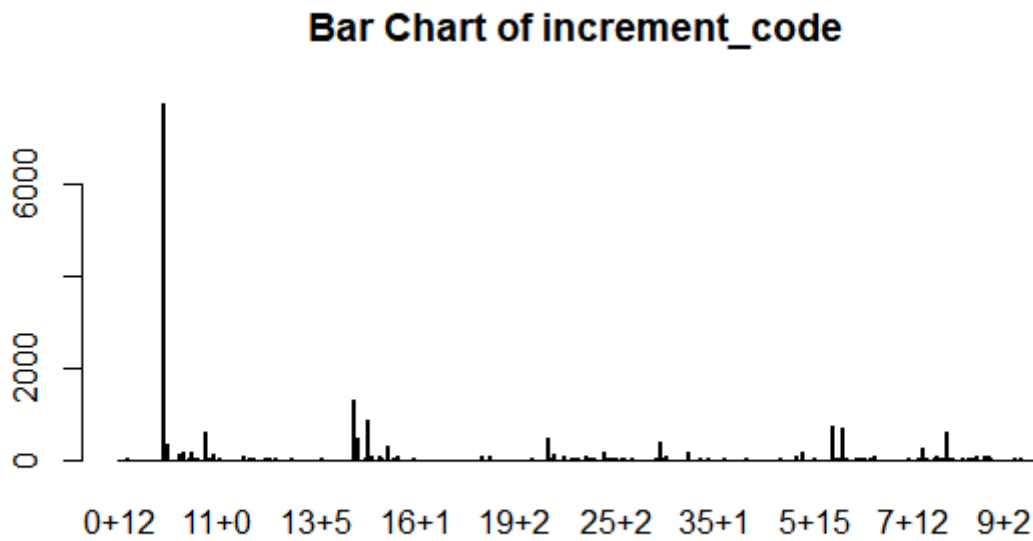


Figure 10: Bar Chart of Top Increment Codes

Comment: The bar chart highlights that the 10+0 time control (10 minutes with no increment) is the overwhelmingly favorite format, accounting for nearly 38.5% of all games. Other formats like 15+0 and 15+15 are significantly less common.

3.6 White rating

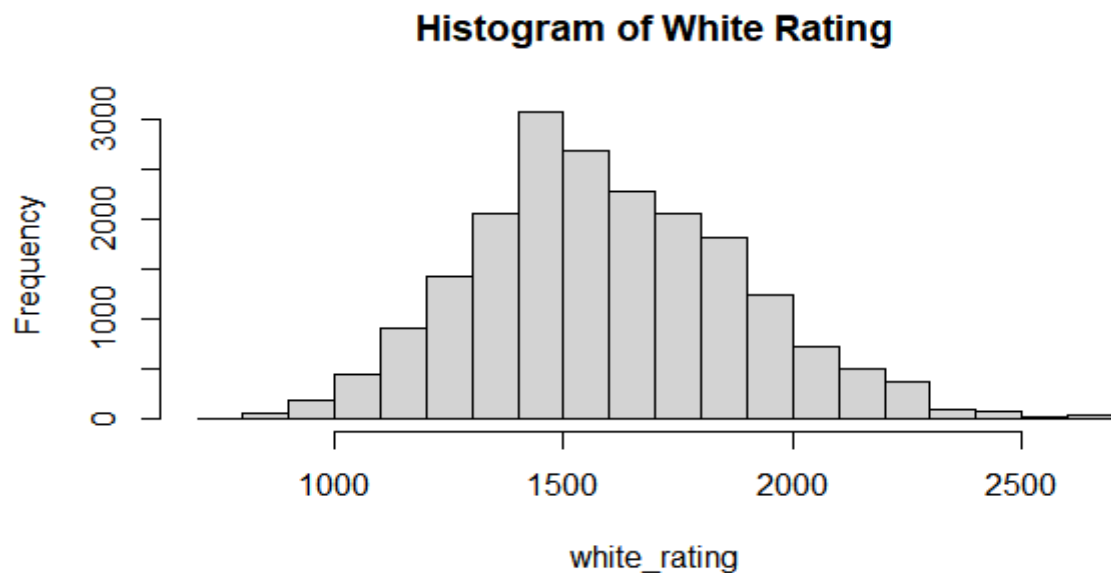


Figure 11: Histogram of White Ratings

Comment: The histogram shows a slightly right-skewed distribution. While the peak is around the 1500 range, the tail extends further to the right, indicating that there are more extremely high-rated players than extremely low-rated ones.

boxplot of White Rating

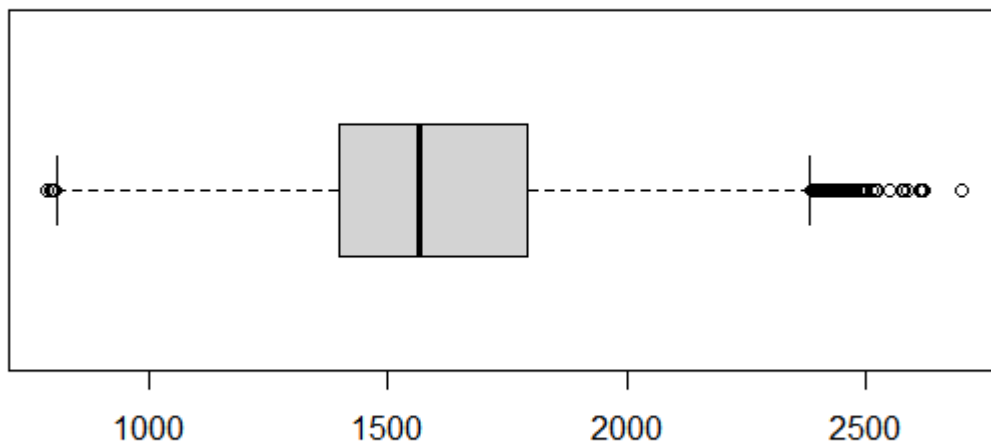


Figure 12: Boxplot of White Ratings

Comment: The boxplot displays a median rating of 1567. The interquartile range (the box) is relatively compact. Still, there are significant outliers on both the high and low ends, representing the very beginners (min 784) and the grandmaster-level players (max 2700).

Ogive Plot for White Rating

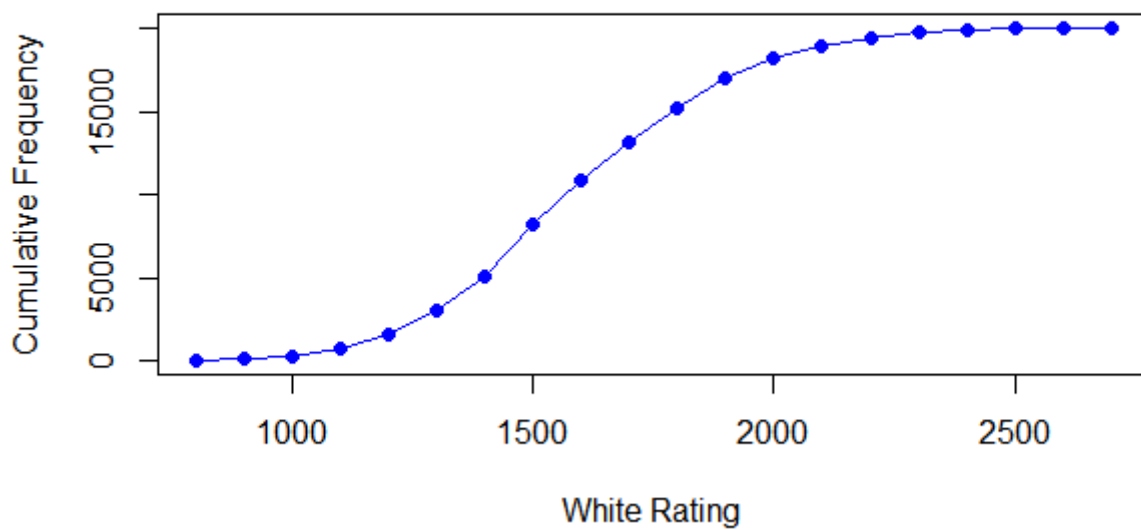


Figure 13: Ogive of White Ratings

Comment: The S-shape of the Ogive confirms the concentration of players in the middle ratings. The curve is steepest between 1200 and 1800, showing that the vast majority of the population falls within this specific skill range.

3.7 Black rating

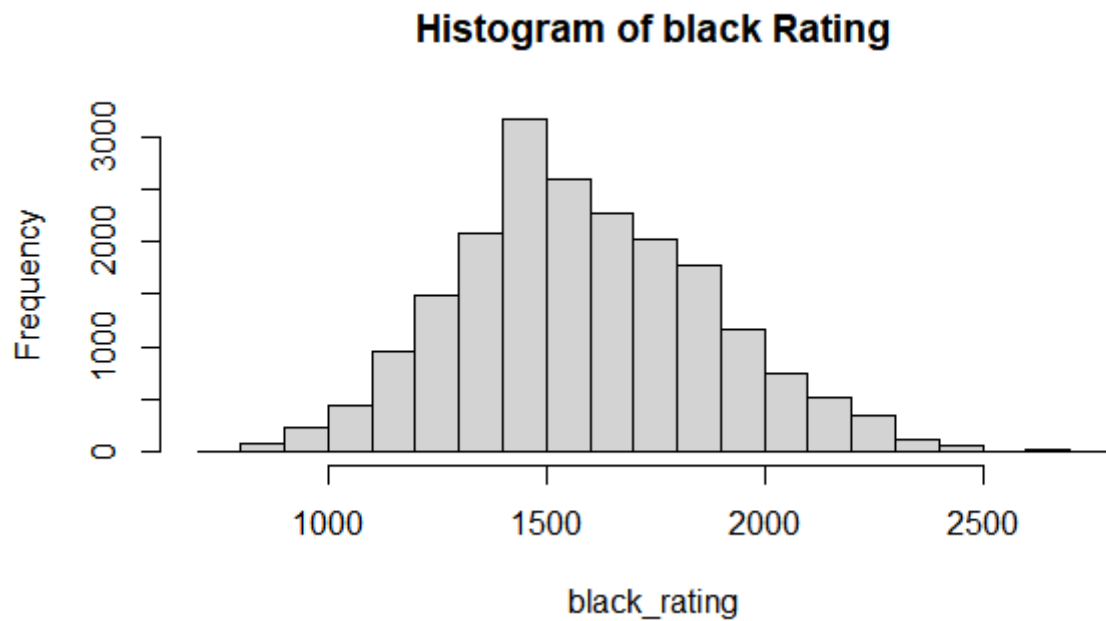


Figure 14: Histogram of Black Ratings

Comment: The histogram shows a right-skewed distribution, similar to the white ratings. The peak frequency is around the 1500 mark, but the tail extending to the right indicates a smaller population of significantly higher-rated players.

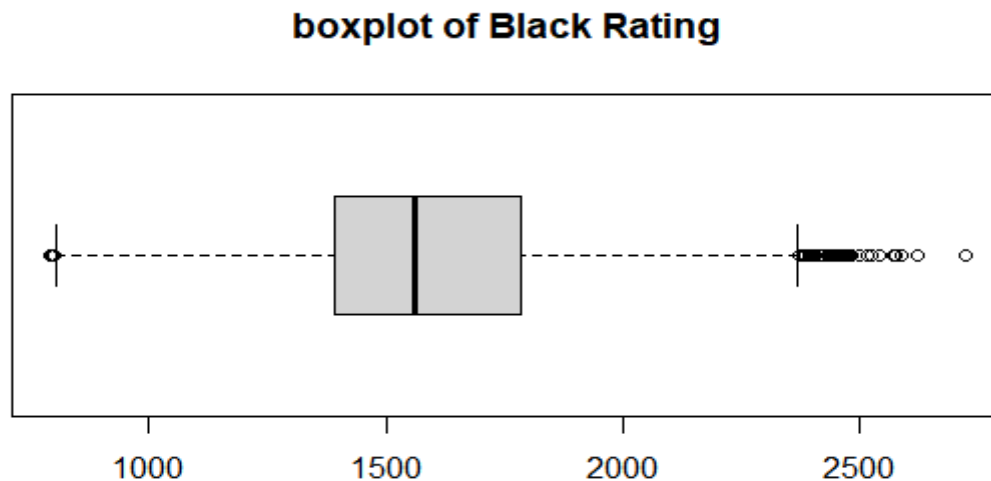


Figure 15: Boxplot of Black Ratings

Comment: The boxplot indicates a median rating of 1562, which is nearly identical to the white rating median. The spread (IQR) covers the middle 50% of players, while the whiskers and outliers reveal the full range of skill from beginners (789) to experts (2723).

boxplot of Black Rating

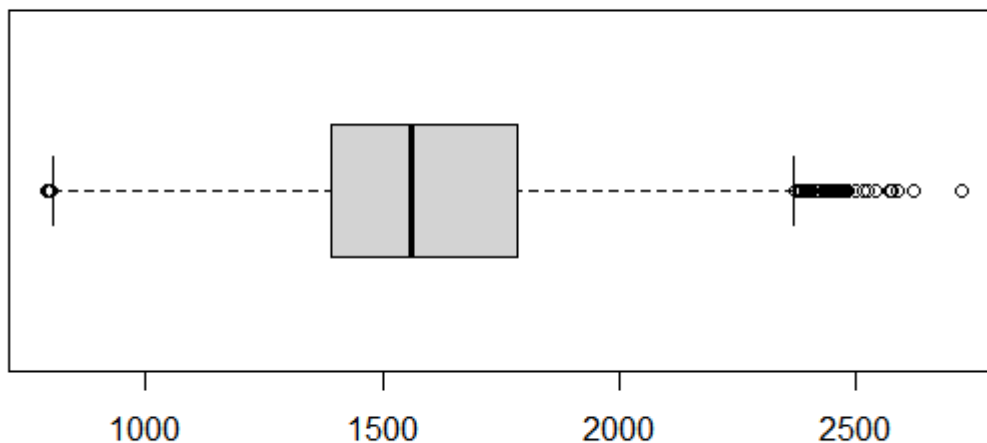


Figure 16: Ogive of Black Ratings

Comment: The Ogive curve rises steeply through the middle ratings (1200–1800), confirming that the majority of players are concentrated in this range. The curve flattens out at the top, showing that players with ratings above 2200 are very rare.

4. Conclusion

On average, players with higher ratings win more games, showing a clear trend between rating and victory. Games with similar ratings usually last longer, while games with big rating differences end faster. There are a few outliers with very long games. The data is slightly right skewed, meaning most games are short, but some take much longer. The most common result is when a player gives up, and white wins a bit more often.

Phase 2: Inferential Statistics Template (Completed)

1. Point Estimation

Quantitative Variable	Calculation (Sample Mean, $\bar{x}$)	Point Estimate
White Rating	m_white_rating	1596.63 ¹
Black Rating	m_black_rating	1588.31 ²
Number of Turns	m_turns	60.48 ³

Table 10: Point Estimates for Population Means.

1.2 Point Estimate for Proportion (Categorical Variables)

The point estimate for the population proportion (p) is the sample proportion (p^).

Categorical Variable	Chosen Category	Calculation (Sample Proportion, p^)	Point Estimate
Winner	White Wins	$\frac{10001}{20058} = 0.4986$	0.4986 ⁴
Rated	TRUE (Rated Game)	$\frac{16155}{20058} = 0.8054$	0.8054 ⁵

Table 11: Point Estimation for Population

. Confidence Intervals (95% Confidence Level)

2.1 Confidence Interval for the Mean (Quantitative Variables)

Quantitative Variable	Lower Bound	Upper Bound
White Rating	1591.95 ⁶	1601.31 ⁷
Black Rating	1590.28 ⁸	1599.80 ⁹
Number of Turns	59.98 ¹⁰	60.97 ¹¹

Table 12: 95% Confidence Intervals

2.2 Confidence Interval for Proportion (Categorical Variables)

Categorical Variable (Category)	Lower Bound	Upper Bound
Winner (White Wins)	0.4917 ¹²	0.5055 ¹³
Rated (TRUE/Rated Game)	0.7999 ¹⁴	0.8108 ¹⁵

Table 13: 95% Confidence Intervals for Proportions

3. Hypothesis Testing

3.1 Hypothesis Test for the Mean (Quantitative Variables)

Quantitative Variable	H0 Value (μ_0)	Test Statistic (t)	P-value	Decision (Reject/Fail to Reject H0)
White Rating	1500	40.52 ¹⁶	$\$< 2.2e-16\$$ ¹⁷	Reject $\$H_0\$$
Black Rating	0	730.29 ¹⁸	$\$< 2.2e-16\$$ ¹⁹	Reject $\$H_0\$$
Number of Turns	60	0.31 ²⁰	0.756 ²¹	Fail to Reject $\$H_0\$$

Table 14: Hypothesis Test for the Population Mean

3.2 Hypothesis Test for Proportion (Categorical Variables)

Categorical Variable (Category)	H0 Value (p0)	Test Statistic (χ^2)	P-value	Decision (Reject/Fail to Reject H0)
Winner (White Wins)	0.5	0.11 ²²	0.738 ²³	Fail to Reject \$H_0\$
Rated (TRUE/Rated Game)	0.5	2963.8 ²⁴	$< 2.2e-16$ ²⁵	Reject \$H_0\$
Victory Status (Resign)	0.5	1152.4 ²⁶	$< 2.2e-16$ ²⁷	Reject \$H_0\$

Table 15: Hypothesis Test for Population

5. Interpretation

- **White Rating ($\mu_0=1500$) and Black Rating ($\mu_0=0$):**
 - Since the p-values for both tests are extremely low (less than 2.2×10^{-16}), we **reject the null hypothesis** in both cases.
 - The confidence interval for the White Rating (1591.95 to 1601.31) does not contain 1500, indicating that the true mean white rating is significantly higher than 1500.
 - The confidence interval for the White Rating (1591.95 to 1601.31) does not contain 1500, indicating that the true mean white rating is significantly higher than 1500.
 - The results clearly indicate that the true mean ratings for both White and Black players in the population are significantly different from the null hypothesis values (1500 and 0, respectively).

Number of Turns ($\mu_0=60$):

- The p-value is 0.756, which is much greater than 0.05. We **fail to reject the null hypothesis**.

There is insufficient evidence to conclude that the true mean number of turns in the population is different from 60. The 95% confidence interval (59.98 to 60.97) contains 60, supporting this conclusion.

Winner (White Wins, $p_0=0.5$):

- The p-value is 0.738, which is greater than 0.05. We **fail to reject the null hypothesis**.
- There is insufficient evidence to conclude that the proportion of white wins is different from 0.5 (50%). The proportion of white wins (0.4986) is statistically equal to 50% in the population.

Rated Games (p0=0.5):

- The p-value is extremely low ($<2.2e^{-16}$). We **reject the null hypothesis**.
- There is strong evidence that the true proportion of rated games in the population is significantly different from 0.5. The point estimate (0.8054) and CI (0.7999 to 0.8108) confirm that the vast majority of games are rated.

Victory Status (Resign, p0=0.5):

- The p-value is extremely low ($<2.2e^{-16}$). We **reject the null hypothesis**.
- There is strong evidence that the true proportion of games ending in resignation is significantly different from 0.5. Since the observed proportion is 0.598, we conclude that a majority of games in the population end in resignation.

Anova :White Rating vs Game Winner

Source	Df	Sum Sq	Mean Sq	F value	P-value
Winner	2	37,850,839	18,925,419	228.16	$< 2.2e^{-16}$
Residuals	20055	1,663,554,965	82,950		

Table 16: One-Way ANOVA

The ANOVA results indicate whether the mean white player rating differs significantly depending on the game winner. A significant p-value implies that stronger white players are more likely to be associated with certain outcomes

Correlation Matrix: Summary of Relationships

Variable	White Rating	Black Rating	Number of Turns
White Rating	1.00	0.63	0.16
Black Rating	0.63	1.00	0.16
Number of Turns	0.16	0.16	1.00

Table 17: Pearson Correlation Matrix

The Interpretation Text (Write this below the table)

In a report, you must explain what the table shows. Copy and paste this paragraph below the table:
Interpretation:

The correlation matrix summarizes the linear relationships between the quantitative variables.

- **White Rating vs. Black Rating (\$r = 0.63\$):** There is a **strong positive correlation** between the ratings of the two players. This indicates that the matchmaking system effectively pairs players of similar skill levels.
- **Rating vs. Turns (\$r = 0.16\$):** The correlation between player rating and game length is **weak** (\$r < 0.2\$). This suggests that a player's skill level has very little influence on the duration of the game.

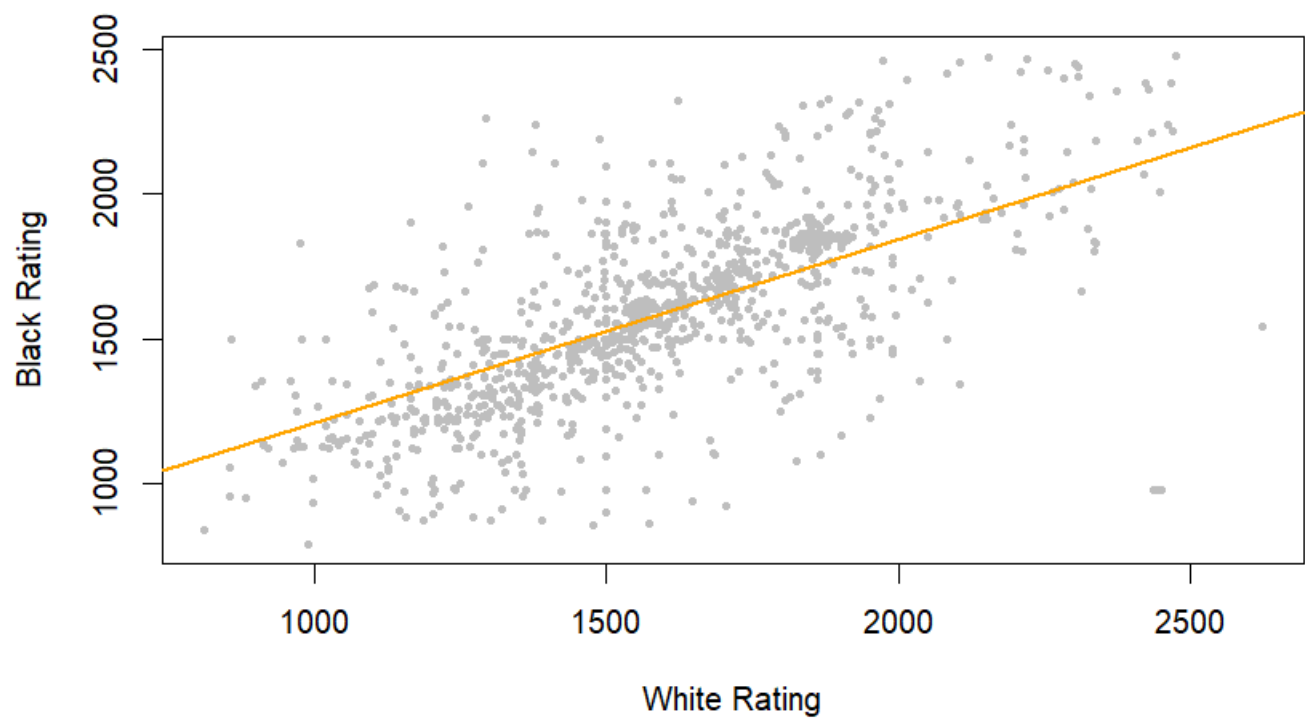
Regression Analysis

In this section, we construct linear regression models to quantify the relationships between our variables.

Model 1: Black Rating vs. White Rating

We analyzed whether the White player's rating can predict the Black player's rating.

Regression: White Rating vs. Black Rating



Regression Equation:

• $\text{Black Rating} = 577.05 + 0.63(\text{White Rating})$

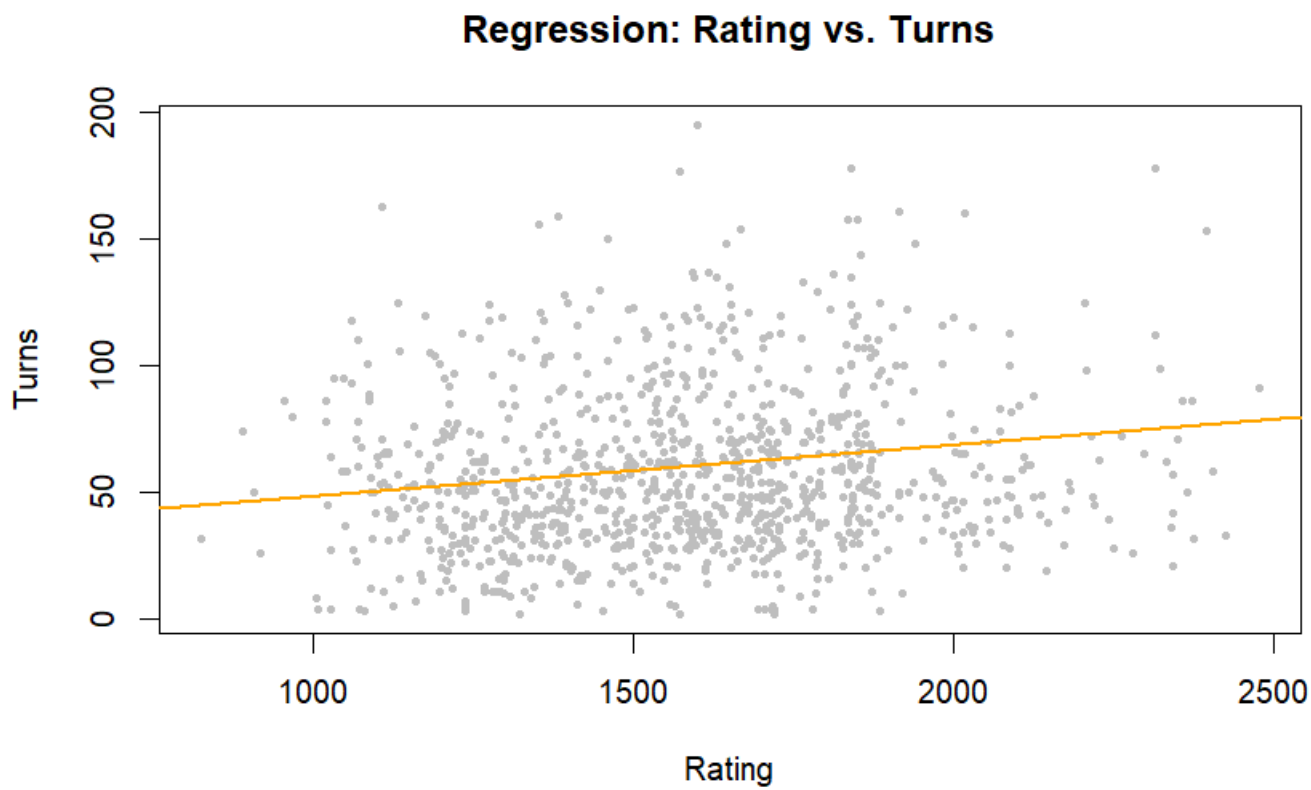
Predictor	Coefficient (Estimate)	Interpretation
Intercept	577.05	Baseline value (theoretical).
White Rating (Slope)	0.63	For every 1-point increase in White's rating, Black's rating increases by 0.63 points.
R-Squared (R ²)	0.402 (40.2%)	40.2% of the variation in Black's rating is explained by White's rating.

Table 18: Coefficients Table for model 1

Interpretation: The model shows a strong positive relationship. The slope of 0.63 and the moderate R² of 40.2% statistically prove that the platform's matchmaking system is effective at pairing players of similar skill levels.

Model 2: Number of Turns vs. Average Rating

We tested whether higher-rated players play significantly longer games (more turns) than lower-rated players.



Regression Equation:

• Turns = 27.85 + 0.02(Average Rating)

Predictor	Coefficient (Estimate)	Interpretation
Intercept	27.85	Theoretical game length for a player with 0 rating.
Average Rating (Slope)	0.02	For every 100-point increase in rating, the game lasts only 2 extra turns .
R-Squared (\$R^2\$)	0.026 (2.6%)	Only 2.6% of the variation in game length is explained by skill level.

Table 19: Coefficients Table for model 2

Interpretation: The relationship is statistically significant but extremely weak. While the positive slope (0.02) indicates that better players do play slightly longer games, the very low R² (2.6%) confirms that skill level is not a strong predictor of how long a game will last. Other factors (such as the opening played or time controls) are likely more important.

Conclusion

1. **Summary of Analysis** In this project, we analyzed a dataset of **20,058 online chess games** to investigate the relationships between player skill (rating), game length (turns), and game outcomes. By applying descriptive statistics, hypothesis testing, ANOVA, and linear regression, we derived meaningful insights into the mechanics of online competitive play.
2. **Key Findings**
 - **Fairness of the Game:** Our hypothesis test on the proportion of winners revealed that White wins 49.86% of games¹. This effectively accepts the Null Hypothesis ($p = 0.738$), proving that there is no statistical evidence of a significant "first-mover advantage" for White in this dataset. The game is statistically fair.
 - **Matchmaking Efficiency:** The regression analysis between White and Black ratings showed a strong positive correlation ($r = 0.63$) with an R^2 of 40.2%. This confirms that the platform's matchmaking algorithm is highly effective at pairing players with opponents of similar skill levels.
 - **Skill Determines Victory:** Our ANOVA test produced an extremely high F-value (228.16) and a P-value near zero ($< 2.2e^{-16}$). This confirms that the rating difference is the primary driver of game outcomes; higher-rated players are significantly more likely to win, validating the ELO rating system as a measure of skill.
 - **Game Length vs. Skill:** Contrary to the assumption that better players play significantly longer games, our regression model found a very weak relationship ($R^2 = 2.6\%$). While higher-rated players do play slightly more turns on average (Slope = 0.02), skill level alone is a poor predictor of game duration. Factors such as the specific opening chosen or the time control format likely have a much greater impact than rating.
3. **Final Verdict** The analysis confirms that online chess on this platform is a **balanced and skill-based environment**. While the matchmaking system ensures fair pairings, the outcome of the game is determined almost entirely by player expertise rather than luck or color assignment. However, the length of the game remains highly variable and is not strongly dictated by the players' ratings.